

PROJECT PROPOSAL

A. GENERAL INFORMATION

A1. Project Title

"Digital Twin Reconstruction of Base Transceiver Stations with 3D Gaussian Splatting: Studying Vanilla 3DGS and Building an Improved 3DGS Variant Better Suited to BTS Towers"

A2. Deadline

Monday 30/11/2026

A3. Principal Investigator

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University: International University – Vietnam National University Ho Chi Minh City

A4. Project Members (including the Principal Investigator)

No.	Full name	Student ID	University / School
1	Lê Anh Kiệt	ITCSIU24047	International University – VNU-HCM

B. RESEARCH DESCRIPTION

B1. Introduction

This is a course project. Its goal is to study how the original (vanilla) 3D Gaussian Splatting (3DGS) pipeline works, use it to reconstruct a real Base Transceiver Station (BTS) telecom tower as a digital twin from drone photographs, and look for a rendering method that works better than vanilla 3DGS for this kind of scene. A BTS scene is mostly flat surfaces (roofs, walls, sky), but the tower mast, antennas, and cables are thin structures that are harder to render well.

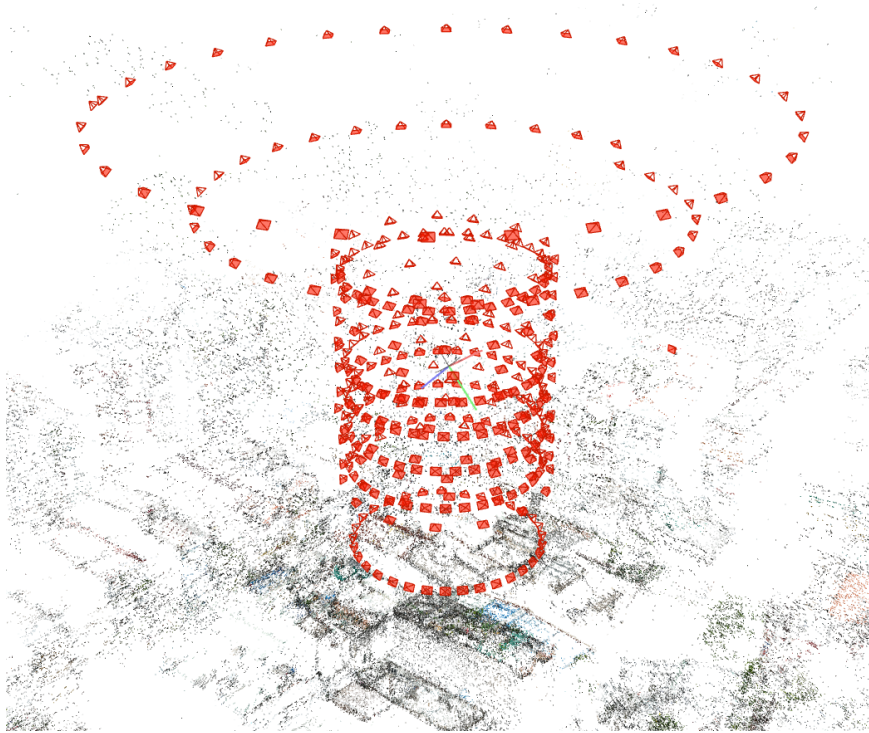


Figure 1: "Camera poses and sparse point cloud generated from the BTS tower dataset."

B2. Objectives and Research Content

- Study the original 3DGS pipeline and its training/density-control bottlenecks as the baseline.
- Survey multiple published techniques for faster training and better densification, and verify each one against its own source code. We will focus on techniques that improve rendering of thin structures (the tower mast, antennas, cables), such as anti-aliasing, opacity/depth regularization, and density control for sparse regions.
- Select and combine the most effective techniques into one optimized pipeline for BTS digital-twin reconstruction.



Figure 2: Output example: a BTS tower rendered by the trained 3DGS model from a new camera view.

- Benchmark the combined pipeline against vanilla 3DGS using PSNR, SSIM, LPIPS, Gaussian count, training time, and memory use.

B3. Expected Results

- A short, verified report and slides comparing multiple 3DGS optimization techniques.
- An optimized 3DGS pipeline for BTS digital twins, combining the best-performing techniques.
- A real benchmark table comparing the optimized pipeline with vanilla 3DGS.

B4. Why this project is suitable for a Computer Graphics course project

- **T1 - Transformations/Hierarchy (Yes - Extensive):** Each Gaussian point in 3D space is defined by geometric transformations: Position (Translation), Rotation (often using Quaternions), and Scale. Additionally, to rapidly render millions of points, 3DGS employs spatial sorting/hierarchical structures (such as depth-based Radix Sort) instead of traditional Hierarchy structures.
- **T2 - Camera/Projection (Yes - Core):** The term “Splatting” refers exactly to this process. 3DGS uses Camera matrices (View & Projection matrices) to project these 3D Gaussians onto the 2D screen plane, then rasterizes them into pixels based on the viewing angle.
- **T4 - Lighting/Shadows (Indirect / Baked):** The original 3DGS does not compute dynamic lighting or shadows. Lighting, shadow, and reflection effects (view-

dependent specular) are all “baked” into the Spherical Harmonics parameters during the training process. The scene’s luminance changes as you shift your viewing angle, but no actual light source is dynamically created.

- **T6 - Interaction/GUI (Yes):** The biggest advantage of 3DGS is its real-time rendering speed (extremely high FPS). This enables the creation of GUIs/Viewers that allow users to interact directly with the scene (rotate, zoom, and pan the camera) with incredible smoothness.